Database Systems Assignment 1

Q1: Hospital Management System

Entities:

- Doctor → DoctorID
- Department → DepartmentID
- Patients → PatientID
- Insurance → InsuranceID
- Appointments → AppointmentID
- Treatments → TreatmentID
- TreatmentProcedure → Weak Entity

Relationships:

- Doctor - Specialty (Primary) → 1:1, Total on Doctor side, Partial on Specialty side.
- Doctor - Specialty (Additional) → 1:M, Partial on both sides.
- Doctor - Department → M:N, Total on both sides.
- Patient - Insurance → 1:M, Partial on Patient side, Total on Insurance side.
- Appointment - Doctor → M:1, Total on Appointment side, Partial on Doctor side.
- Appointment - Patient → M:1, Total on Appointment side, Partial on Patient side.
- Appointment - Department → 1:1, Total on Appointment side, Partial on Department side.
- Appointment - Treatment → 1:M, Total on Treatment side, Partial on Appointment side.
- Treatment - Procedure → M:N, Partial on both sides.

- Treatment - FollowUp → 1:M, Total on FollowUp side, Partial on Treatment side.

Attributes:

Doctor

- DoctorID
- FullName (Composite: FirstName + LastName)
- PrimarySpecialty
- AdditionalSpecialty (Multi-Valued)
- ContactInformation (Composite: PhoneNumbers + Email, PhoneNumber is Multi-Valued)

Department

- DepartmentID
- DepartmentName
- DepartmentLocation

Patient

- PatientID
- FullName (Composite: FirstName + LastName)
- DateOfBirth
- HomeAddress

Insurance

- InsuranceID
- ProviderName
- CoverageDetails
- PatientID (Foreign Key)

Appointment

- AppointmentID

- Status
- ReasonForVisit
- Timestamp
- RoomNumber
- DepartmentID (Foreign Key)

Treatment

- TreatmentID
- Description
- Cost
- TreatmentDate
- AppointmentID (Foreign Key)

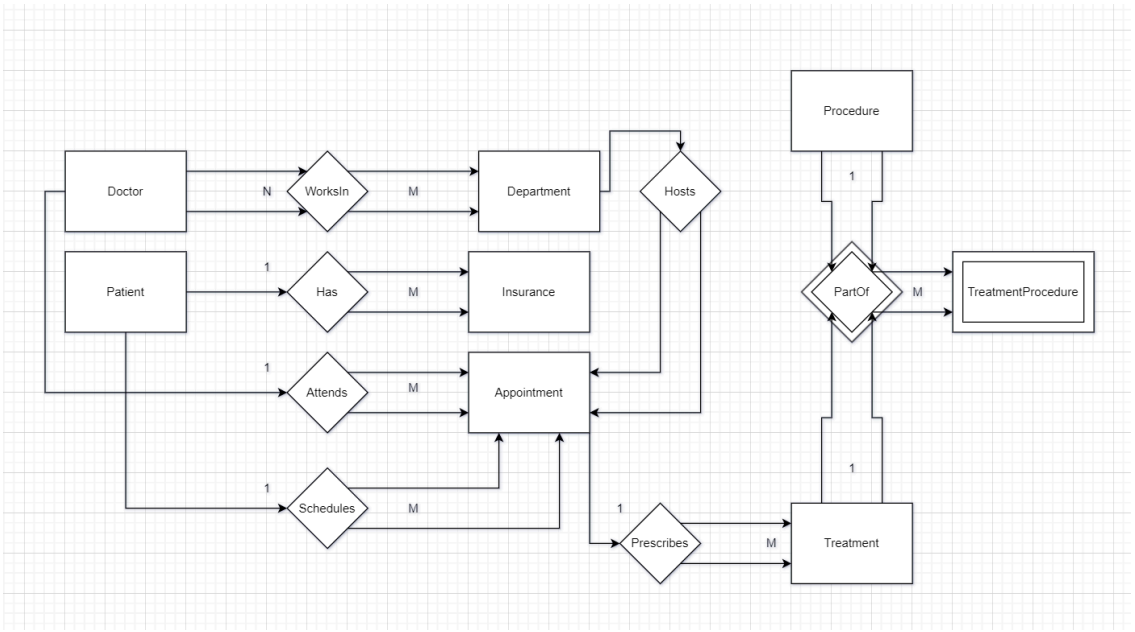
Procedure

- ProcedureID
- ProcedureName
- Duration
- RequiredEquipment (Multi-Valued)

TreatmentProcedure (Weak Entity)

- FollowUpActions

ER Diagram:



Q2: Smart Transportation System

Entities and Primary Keys:

Vehicle System

- Vehicle (PK: VehicleID)
- Maintenance (PK: MaintenanceID)
- ChargingStation (PK: StationID)
- Sensor (PK: SensorID)
- MonitoringSystem (PK: SystemID)

Passenger Operations

- User (PK: UserID)
- Booking (PK: BookingID)
- Payment (PK: PaymentID)
- Feedback (PK: FeedbackID)
- Subscription (PK: SubscriptionID)

Infrastructure Components

- TrafficSignal (PK: SignalID)
- ParkingZone (PK: ZoneID)
- RoadConditionMonitor (PK: MonitorID)
- LaneManagement (PK: LaneID)

Safety & Monitoring

- Incident (PK: IncidentID)
- EmergencyService (PK: ServiceID)
- Alert (PK: AlertID)
- SafetyInspection (PK: InspectionID)

Special Considerations

- DynamicPricing (PK: PricingID)
- ServiceProvider (PK: ProviderID)

Relationships and Cardinalities:

Vehicle System

- Vehicle - Maintenance → 1:M, Total on Maintenance
- Vehicle - ChargingStation → M:N, Partial on both
- Vehicle - Sensor → 1:M, Total on Both
- Vehicle - MonitoringSystem → 1:M, Total on MonitoringSystem
- Vehicle - Vehicle → M:N, Partial on both

Passenger Operations

- User - Booking → 1:M, Total on Booking
- User - Payment → 1:M, Total on Payment
- User - Feedback → 1:M, Total on Feedback
- User - Subscription → M:N, Partial on both
- Booking - Feedback → 1:1, Total on Feedback

Infrastructure Components

- Vehicle - TrafficSignal → M:N, Partial on both
- Vehicle - ParkingZone → M:N, Partial on both
- Vehicle - RoadConditionMonitor → M:N, Partial on both
- Vehicle - LaneManagement → M:N, Partial on both

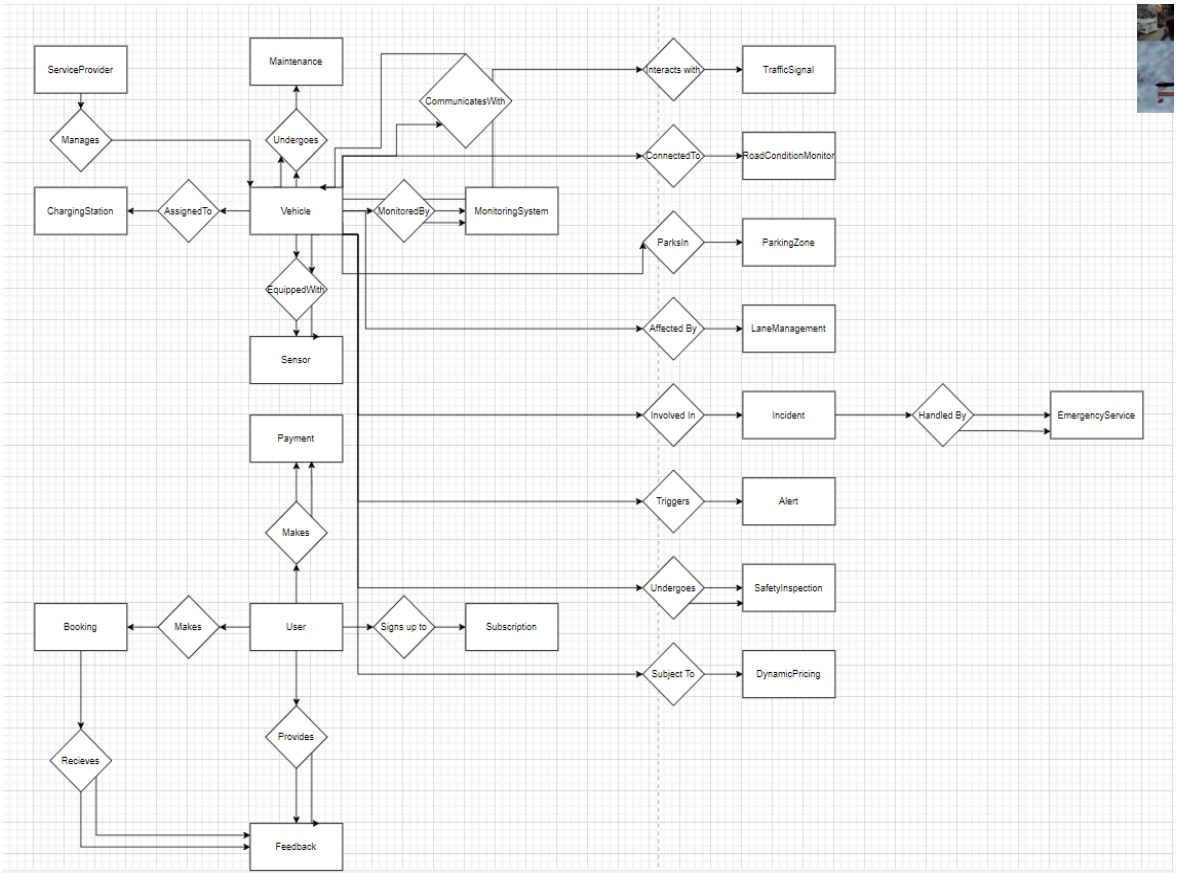
Safety & Monitoring

- Vehicle - Incident → M:N, Partial on both
- Incident - EmergencyService → 1:M, Total on EmergencyService
- Vehicle - Alert → M:N, Partial on both
- Vehicle - SafetyInspection → 1:M, Total on SafetyInspection

Special Considerations

- Vehicle - DynamicPricing → M:N, Partial on both
- ServiceProvider - Vehicle → M:N, Partial on both

ER Diagram:



Q3: FYP Management System

Entities:

- Student → StudentID
- Group → GroupID
- Project → ProjectID
- Faculty → FacultyID
- Evaluation → EvaluationID
- ProjectVersion → VersionID
- Feedback → FeedbackID
- EvaluationScore → Weak Entity (EvaluationID, FacultyID)

- Supervision → Weak Entity (ProjectID, FacultyID)

Relationships:

- Student - Group → N:1, Total on both
- Group - Project → 1:1, Total on both
- Project - Faculty (Supervision) → M:N, Total on Project side, Partial on Faculty side
- Project - Evaluation → 1:M, Total on Evaluation side, Partial on Project side
- Evaluation - Faculty (Evaluators) → M:N, Partial on both sides
- Project - ProjectVersion → 1:M, Partial on Project side
- ProjectVersion - Feedback → 1:M, Partial on ProjectVersion side
- Feedback - Faculty → M:1, Total on Feedback side

Attributes:

Student

- StudentID
- FullName (Composite: FirstName + LastName)
- Email
- Contact
- Department

Group

- GroupID
- ProjectTitle

Project

- ProjectID
- Title
- Description

Faculty

- FacultyID
- FullName (Composite: FirstName + LastName)
- Email
- Contact
- Department

Evaluation

- EvaluationID
- Phase (Proposal/Mid-Term/Final)
- Date
- Status

ProjectVersion

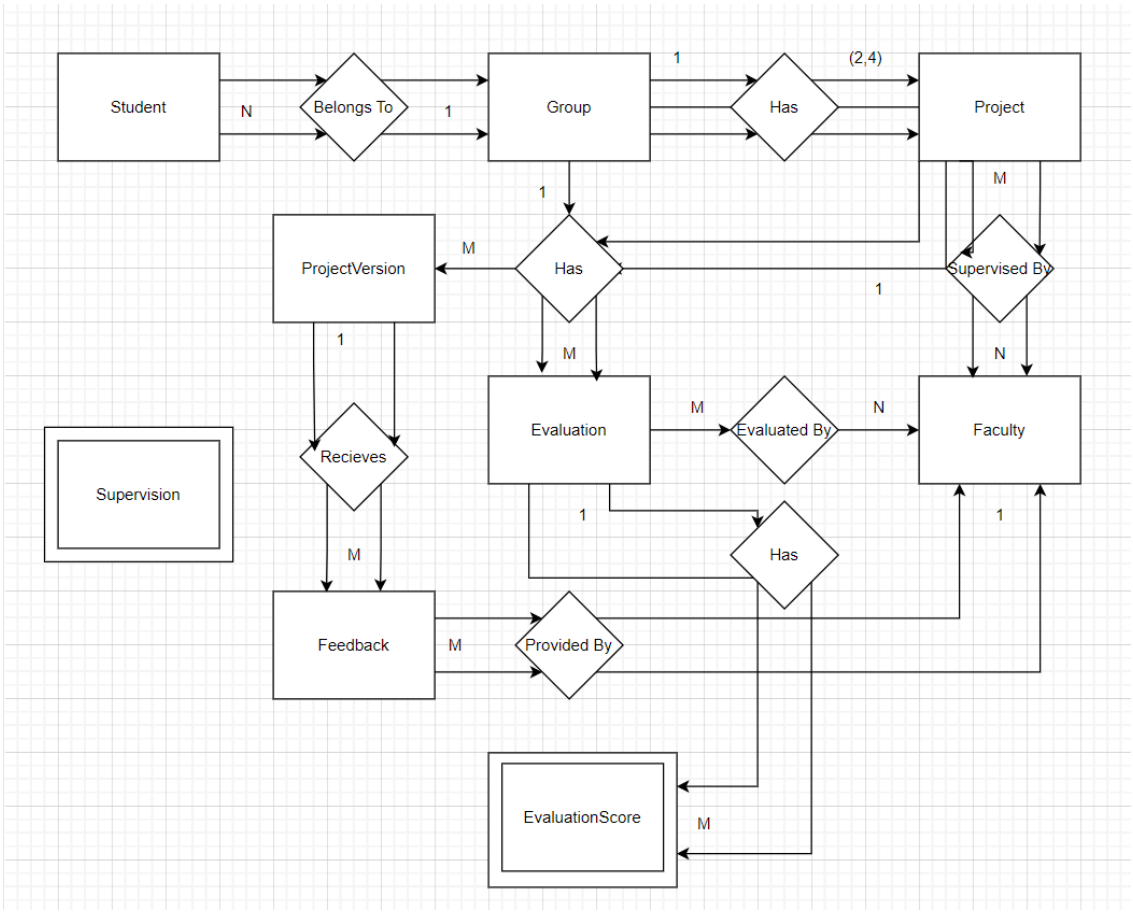
- VersionID
- ModificationDate
- Changes
- Status

Feedback

- FeedbackID
- Comments
- Timestamp

EvaluationScore (Weak Entity)

- EvaluationID
- FacultyID
- Score (0–100)
- Comment


Student
N
Belongs To
1
Group
1
Has
(2,4)
Project
1
M
ProjectVersion
M
Has
Supervised By
1
1
M
N
Evaluation
M
Evaluated By
N
Faculty
Supervision
Recieves
1
1
M
Has
Feedback
M
Provided By
EvaluationScore
M